

Title: DPT

Date:

dynamics

 α : constant growth

 β : high order

 Δ : triangle

 z : $\langle k \rangle$

$$\frac{dz}{dt} = \alpha - rz + \beta(z^2 - \Delta)$$

$$W[A_{ij}(0 \rightarrow 1)] = \left[\frac{\alpha}{N} + \beta(A^2)_{ij} \right]$$

$$W[A_{ij}(1 \rightarrow 0)] = r$$

$$\begin{aligned} \frac{dz}{dt} &= \alpha \left(1 - \frac{z}{N} \right) - rz \\ &+ \frac{\beta}{N} \left(\sum_{ijk} A_{ik} A_{kj} (1 - A_{ij}) \right) z^2 - \Delta \end{aligned}$$

hyperbolic scaling

$$z(t) \sim \begin{cases} \text{constant} & (\beta \leq \frac{r^2}{4\alpha}, t \rightarrow \infty) \\ 1/(t - t_c) & (\beta > \frac{r^2}{4\alpha}, t \rightarrow t_c) \end{cases}$$

$$\begin{aligned} t_c &= w^{-1} [\text{acot}(2w) + \pi/2] \\ (w &= \sqrt{|\alpha\beta - r^2/4|}) \end{aligned}$$

$$\frac{dz}{dt} = \alpha - rz + \beta z^2$$

$$1/(\alpha - rz + \beta z^2) dz = dt$$

$$\frac{1}{pu^2 + q} du = dt$$

$$p = \beta, u = z - \frac{r}{2\beta}, q = \alpha - \frac{r^2}{4\beta}$$

$$\frac{1}{\sqrt{pq}} \text{atan} \sqrt{\frac{p}{q}} u = t + C$$

$$t=0, z=0: C = \frac{1}{\sqrt{\alpha\beta - \frac{r^2}{4}}} \text{atan} \frac{-r}{\sqrt{4\alpha\beta - r^2}}$$

$$w = \sqrt{pq} \quad C = w^{-1} \text{atan} -\frac{r}{2w}$$

$$\begin{aligned} t_c &= -C + \frac{\pi}{2} w^{-1} \\ &= w^{-1} \left(\frac{\pi}{2} + \text{acot} \frac{2w}{r} \right) \end{aligned}$$

Title: DPT

Date:

after DPT

P : proportion of nodes
with no connection to core

$$q: \langle k \rangle / N$$

$$q(t) = 1 - P(t)^2$$

$$dp/dt = \alpha t$$

$$P(t) = P_0 e^{-\alpha(t-t_0)}$$

adiabatic approx:

$$\frac{d\Delta}{dt} = \beta(z^2 - \Delta) - 3r\Delta + O(\beta^2)$$

$$\frac{d\Delta}{dt} = 0, \Delta = \frac{\beta z^2}{3r + \beta}$$

$$\beta' z^2 = \beta(z^2 - \Delta)$$

$$\beta' = \beta - \frac{\beta^2}{3r + \beta}$$

$$= \beta - \frac{\beta^2}{3r} \sum_{i=0}^{\infty} \left(-\frac{\beta}{3r}\right)^i$$

Master eq.

$$G(x, t) = \sum_k P(k, t) x^k$$

$$\begin{aligned} \frac{\partial}{\partial t} P(k, t) = & [\alpha + \beta(k-1)z(t)] P(k-1, t) \\ & - [\alpha + \beta k z(t) + r k] P(k, t) \\ & + r(k+1) P(k+1, t) \end{aligned}$$

$$k P(k, t) = x \cdot \frac{\partial G}{\partial x}$$

$$\begin{aligned} \cdot x^k : \frac{\partial G}{\partial t} = & (\alpha x - \alpha) G + (-r x + r) \frac{\partial G}{\partial x} \\ & (\beta z x^2 - \beta z x) \frac{\partial G}{\partial x} \end{aligned}$$

$$\frac{\partial G}{\partial t} = (\beta z x - r)(x-1) \frac{\partial G}{\partial x} + \alpha(x-1) G$$